

## Research Interests

- Graph Theory & Graph Algorithms
- Design & Analysis of Algorithms
- Community Detection
- Social Network Analysis
- Complex Networks
- Machine Learning on Graphs

## Education

- **Concordia University** Sep 2024 – 2027 (expected)  
Master of Computer Science (Thesis-based), advised by [Prof. Hovhannes Harutyunyan](#)  
Thesis: *Geodesic-Compact Community Detection and Quality Assessment in Social Networks*  
Graduate Coursework: Advanced Analysis of Algorithms, Combinatorial Algorithms, Algorithm Design Techniques, Machine Learning  
Montréal, QC, Canada
- **Vali-e-Asr University of Rafsanjan** Sep 2018 – Jun 2023  
B.Sc. in Computer Engineering, advised by [Dr. Fahimeh Dabaghi-Zarandi](#)  
*Graduated with Honors* | B.Sc. Thesis Grade: **20/20**  
Thesis: *Community detection in complex networks via an improved randomized algorithm* → published in the *Journal of Network and Computer Applications* (Q1)  
Rafsanjan, Iran

## Publications

### Journal Articles

- [1] *Spider Community Detection: Seeded Geodesic Expansion with Modularity-Guided Refinement and Greedy Merge Matching*. H. Harutyunyan, **P. Kamalipour** (2026). *Computers*, 15(2), 83. [\[DOI\]](#)
- [2] *Community Detection in Complex Networks Based on an Improved Random Algorithm Using Local and Global Network Information*. F. Dabaghi-Zarandi, **P. Kamalipour** (2022). *Journal of Network and Computer Applications* (Q1), vol. 206. [\[DOI\]](#) → Cited 25+ times.

### Peer-Reviewed Conference Papers

- [3] *From Dense Graphs to Meaningful Communities: Assessing Community Quality Using Geodesic Distance Modularity on Metric Backbone-Sparsified Networks*. **P. Kamalipour**, H. Harutyunyan (2025). *IEEE SNAMS 2025*, Vienna, Austria, pp. 234–241. [\[DOI\]](#)
- [4] *Extending Refactoring Detection to Kotlin: A Dataset and Comparative Study*. I. Hemati Moghadam, M. M. Afkhami, **P. Kamalipour**, V. Zaytsev (2024). *IEEE SANER 2024*, Rovaniemi, Finland, pp. 267–271. [\[DOI\]](#)

### Under Review

- [5] *XlateRefactor: A Language-Agnostic Pipeline for Cross-Language Refactoring Detection Using LLM-Based Code Translation*. I. Hemati Moghadam, M. M. Afkhami, V. Zaytsev, M. H. Ashoori, H. Bazmandegan, F. Zarepour, **P. Kamalipour**. Under review at JSME (major revision).

## Research Experience

- **Graduate Research Assistant** – Concordia University Aug 2024 – Present  
[Algorithms & Complexity Lab](#), Advisor: Prof. Hovhannes Harutyunyan  
Montréal, QC, Canada
  - Designed the **Spider** community detection algorithm, combining seeded geodesic expansion, modularity-guided refinement, and greedy merge matching.
  - Evaluated Spider against **Louvain, Leiden, Infomap, CNM, and Label Propagation** on nine real-world networks (34–8,035 vertices) and LFR benchmarks: top F1 and ARI on Karate Club and Political Blogs, and **NMI 0.88** on Primary School where flow and propagation-based baselines collapse to near-trivial partitions.
  - Introduced **Weighted Average Geodesic Distance Modularity (wGDM)**, a size-consistent, label-free partition quality measure, and conducted the first systematic study of **Metric Backbone** sparsification with Leiden: **14–71%** edge reduction across seven networks while preserving, and on dense contact networks improving, detection quality (High School F1: 0.834 → 0.970).
  - Built a fully reproducible experimental pipeline with fixed random seeds, baseline implementations, and automated evaluation scripts.
- **Research Collaborator (Remote)** – University of Twente Aug 2023 – Mar 2024  
[Formal Methods and Tools \(FMT\) Group](#), Supervisor: Dr. Iman Hemati Moghadam  
Enschede, The Netherlands
  - Implemented the **KotlinCode2Text** parser and integrated it into the **RefDetect** refactoring-detection framework; built two refactoring datasets used in the *SANER 2024* empirical evaluation.
  - Developed and evaluated **LLM prompting strategies** for cross-language code translation, contributing to the **XlateRefactor** pipeline (241 commits, 71 repositories, ~2,000 validated refactorings across Java, Kotlin, and C), with all artifacts released publicly.
- **Undergraduate Research Assistant** – Vali-e-Asr University of Rafsanjan Aug 2021 – Mar 2024  
Department of Computer Engineering, Supervisor: Dr. Fahimeh Dabaghi-Zarandi  
Rafsanjan, Iran
  - Developed the **CRLG** community detection framework based on a randomized algorithm exploiting local and global network information, with weighted probabilistic seeding and heuristic community merging.

- Evaluated on real networks and GN/LFR benchmarks, achieving up to **+10%** over LCDR, MOACO, Node2vec-SC, NE-N2V, CDASS, and TS in NMI, modularity, and density; published in *JNCA* (Q1).

## Teaching Experience

- **Graduate Teaching Assistant** – Concordia University Sep 2024 – Present  
Gina Cody School of Engineering and Computer Science Montréal, QC, Canada  
Recognition: **Dean's Excellence Award for Excellence in Teaching by a Teaching Assistant**, May 2026  
– Accumulated **1,400+** hours of tutorial instruction, laboratory demonstration, grading, and Programmer-On-Duty support across **11 courses** spanning theory, systems, and AI, including:
  - \* COMP 465: Design & Analysis of Algorithms
  - \* COMP 472: Artificial Intelligence
  - \* COMP 339: Combinatorics
  - \* COMP 233: Probability & Statistics for CS
  - \* COMP 335: Intro to Theoretical CS
  - \* SOEN 363: Data Systems for Software Eng.
- **Undergraduate Teaching Assistant** – Vali-e-Asr University of Rafsanjan Mar 2021 – Jan 2024  
Department of Computer Engineering Rafsanjan, Iran  
– Served as **Head TA and tutorial leader** for foundational CS courses (Data Structures, Algorithms Design, Discrete Mathematics, AI, Data Mining), mentoring students and coordinating grading teams.

## Honors & Awards

- **Dean's Excellence Award for Excellence in Teaching by a Teaching Assistant**, Concordia University May 2026
- **Conference and Exposition Allowance**, Concordia University 2025
- **DRW Graduate Scholarship in Computer Science**, Concordia University & DRW Company 2024
- **Concordia Merit Scholarship** (Entrance Scholarship Award), School of Graduate Studies Sep 2024
- **Financial Research Support (FRS)**, Gina Cody School of Engineering and Computer Science 2024
- **Distinguished Student Award** – among all students of Vali-e-Asr University 2023
- **Top Researcher Award** – among all undergraduate students in Kerman Province 2023

## Selected Projects

- **Scalable GNN-Based Community Detection** ([GitHub](#)) 2024  
– Built GCN and GraphSAGE pipelines with neighbor sampling and graph partitioning; reached **90%** accuracy on 10K-node SBM graphs, enabled training on Reddit where full-batch methods run out of memory, and analyzed accuracy–time–memory trade-offs for scalable GNN deployment.
- **Minimum-Cost Flow Algorithms on Randomized Networks** ([GitHub](#)) 2024  
– Implemented Successive Shortest Path from scratch and compared SSP, Capacity Scaling, Scaling-SSP, and Primal–Dual across 28 randomized Euclidean graph configurations; showed Primal–Dual consistently attains optimal cost while SSP degrades with density.

## Conference Presentations

- **IEEE SNAMS 2025 – Vienna, Austria** Nov 2025  
– Oral presentation (in person): *“From Dense Graphs to Meaningful Communities: Assessing Community Quality Using Geodesic Distance Modularity on Metric Backbone-Sparsified Networks.”* [[Slides](#)]

## Academic Service & Leadership

- **Sub-reviewer** for conference submissions, with Prof. Hovhannes Harutyunyan 2025 – Present
- **Director, Research Assistant Committee** – Scientific Association of CE, Vali-e-Asr University 2022 – 2023
- **Director, Teaching Assistant Committee** – Scientific Association of CE, Vali-e-Asr University 2022 – 2023
- **Teaching Staff Member** – Vali-e-Asr Collegiate Programming Contest (VCPC) 2021 – 2022

## Technical Skills

- **Graph Mining & Network Science:** Community Detection, Link Prediction, Node Classification, Network Embeddings, Graph Algorithms, SNAP & LFR Benchmarks, Large-Scale Network Evaluation
- **Programming:** Python, C, C++, C#, Java, MATLAB, Bash, Assembly (x86, ARM)
- **ML & Data:** PyTorch, PyTorch Geometric, Scikit-learn, NetworkX, iGraph, NumPy, Pandas, SciPy, Matplotlib
- **Tools & Databases:** Linux, Git, Docker, L<sup>A</sup>T<sub>E</sub>X, Jupyter, PostgreSQL, MySQL, MongoDB, Neo4j

## Languages

- **English:** Proficient (C1) | **French:** Pre-intermediate (A2, in progress toward B2) | **Persian:** Native

## References

- **Hovhannes Harutyunyan**, Professor, Concordia University [hovhannes.harutyunyan@concordia.ca](mailto:hovhannes.harutyunyan@concordia.ca)  
*Dept. of Computer Science & Software Engineering → Relation: Master's Advisor*
- **Iman Hemati Moghadam**, Researcher, Eindhoven University of Technology [i.hemati.moghadam@tue.nl](mailto:i.hemati.moghadam@tue.nl)  
*Relation: Research Supervisor (SANER 2024 and XlateRefactor studies)*
- **Fahimeh Dabaghi-Zarandi**, Assistant Professor, Vali-e-Asr University of Rafsanjan [f.dabaghi@vru.ac.ir](mailto:f.dabaghi@vru.ac.ir)  
*Dept. of Computer Engineering → Relation: Bachelor's Thesis Advisor*